

Quantile Regression

Mathematical Formulation and Real World Uses

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Table of Contents

Quantile
Regression

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

1 The Concept

2 Quantile Regression

3 MATLAB Solution

4 Results

5 Final Remarks

The Foundation

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Regression

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

In OLS, the parameters (β_0, β_1) are obtained by minimizing the sum of squared residuals, which estimates the conditional mean $E(Y|X)$:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

If we replace the squared loss with the absolute value, we get the LAD, which estimates the conditional median $Q_{0,5}(Y|X)$:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n |y_i - (\beta_0 + \beta_1 x_i)|$$

From Median to Quantiles through optimization

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

The conditional median is a specific quantile; can the loss function be modified to target other quantiles?

The absolute value is symmetric: it penalises positive and negative residuals equally, this is why we get the median. Therefore to target other quantiles, we have to give different weights to positive and negative residuals:

$$\rho_{\tau}(\epsilon) = \begin{cases} \tau\epsilon & \text{if } \epsilon \geq 0 \\ (\tau - 1)\epsilon & \text{if } \epsilon < 0 \end{cases} \quad \text{with } \tau \in (0, 1)$$

where τ is the targeted quantile.

The Optimization Problem

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

Replacing the absolute value with ρ_τ in the regression problem, we obtain the quantile regression, a generalisation of the LAD, which estimates the conditional quantile $Q_\tau(Y|X)$

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n \rho_\tau(y_i - (\beta_0 + \beta_1 x_i))$$

Just like the LAD, this function is piecewise linear and can't be differentiable at 0; one of the approaches is to reformulate and solve it as an LP problem

The LP Reformulation: splitting the residuals

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

Each residual $\epsilon = y_i - (\beta_0 + \beta_1 x_i)$ can be decomposed into its positive and negative parts:

$$\epsilon_i = u_i - v_i, \quad \text{with } u_i, v_i \geq 0$$

$$\left\{ \begin{array}{ll} \epsilon_i \geq 0: & u_i = \epsilon_i, v_i = 0 \\ & \Rightarrow \rho_\tau(\epsilon_i) = \tau \epsilon_i = \tau u_i + (1 - \tau) \cdot 0 \\ \epsilon_i < 0: & u_i = 0, v_i = -\epsilon_i \\ & \Rightarrow \rho_\tau(\epsilon_i) = (\tau - 1) \epsilon_i = \tau \cdot 0 + (1 - \tau) v_i \end{array} \right.$$

In both cases: $\rho_\tau(\epsilon_i) = \tau u_i + (1 - \tau) v_i$

The LP Reformulation: complete problem

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

Therefore the complete reformulation of the problem can be summarized in this form:

$$\begin{aligned} \min_{\beta_0, \beta_1, u_i, v_i} \quad & \tau \sum_{i=1}^n u_i + (1 - \tau) \sum_{i=1}^n v_i \\ \text{s.t.} \quad & y_i - (\beta_0 + \beta_1 x_i) = u_i - v_i \quad \forall i = 1, \dots, n \\ & u_i, v_i \geq 0 \quad \forall i = 1, \dots, n \end{aligned}$$

In this form the problem is an LP, which is computationally trivial, and directly translatable to the solver

YALMIP and Gurobi Implementation

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

```
beta = sdpvar(p, 1);
u = sdpvar(n, 1);
v = sdpvar(n, 1);
objective = tau * sum(u) + (1-tau) * sum(v);
|
options = sdpsettings('solver', 'gurobi', 'verbose', 0);
|
C = [y == X * beta + u - v;
     u >= 0;
     v >= 0];
|
sol = optimize(C, objective, options);
|
if sol.problem ~= 0
|   error("Error! The solver didn't find an optimal solution")
end
|
beta_hat = value(beta);
|
```

Engel Curves via Quantile Regression

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

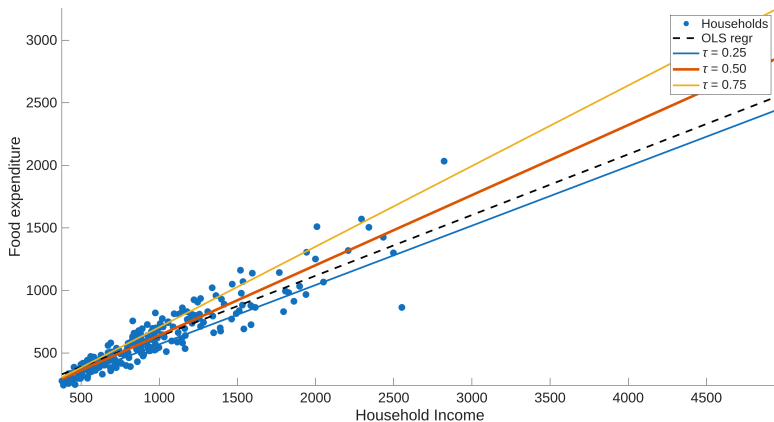


Figure: Engel Curves Dataset (Koenker & Hallock 2001, Fig. 3)

OLS vs Quantile Regression

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

OLS

- Estimates the conditional mean
- Sensitive to outliers (squared loss)
- One optimization problem, one slope
- Can't capture how the relationship varies across the distribution

Quantile Regression

- Estimates any conditional quantile
- Less sensitive to outliers, like LAD
- Family of LPs in τ , multiple slopes
- Naturally reveals how the relationship changes across quantiles

Advantages, Limitations and Applications

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

Advantages

- Less sensitive to outliers than other methods
- Naturally reveals the relationship across different quantiles
- Requires no distributional assumption on the errors

Limitations

- More computationally expensive than linear regression
- The extra robustness is sometimes unnecessary

Applications Suited to any setting where the conditional distribution of $Y|X$ varies in shape across X

- **Demand Analysis:** e.g. Engel Curves (Fig. 1) or Alcohol Demand
- **Labor economics:** e.g. Wage Inequality
- **Medical Research:** e.g. Birthweight (Koenker & Hallock, 2001)

“Just as the mean gives an incomplete picture of a single distribution, so the regression curve gives a corresponding incomplete picture for a set of distributions.”

— Mosteller & Tukey (1977)

“[Quantile regression is] a comprehensive strategy for completing the regression picture.”

— Koenker & Hallock (2001)

References

Quantile
Regression

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The Concept

Quantile
Regression

MATLAB
Solution

Results

Final Remarks

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